

MAT 362 HW01

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Problem. 1.2.2. Let $\alpha(t)$ be a parametrized curve which does not pass through the origin. If $\alpha(t_0)$ is the point of the trace of α which is closest to the origin and $\alpha'(t_0) \neq 0$, show that the position vector $\alpha(t_0)$ is orthogonal to $\alpha'(t_0)$.

Solution. Consider the quantity $\|\alpha(t)\|^2 = \alpha(t) \cdot \alpha(t)$. The length of the vector $\alpha(t)$ is minimized at $t = t_0$, we have that the derivative of $\alpha(t) \cdot \alpha(t)$ at $t = t_0$ is zero, namely

$$2\alpha(t_0) \cdot \alpha'(t_0) = 0.$$

The conditions in the problem guarantee that $\alpha(t_0)$ and $\alpha'(t_0)$ are both not zero vectors, so they must be orthogonal in order for their dot product to be zero. $\square$

Problem. 1.2.4. Let $\alpha: I \rightarrow \mathbb{R}^3$ be a parametrized curve and let $v \in \mathbb{R}^3$ be a fixed vector. Assume that $\alpha'(t)$ is orthogonal to v for all $t \in I$ and that $\alpha(0)$ is also orthogonal to v . Prove that $\alpha(t)$ is orthogonal to v for all $t \in I$.

Solution. We can compute

$$(\alpha(t) \cdot v)' = \alpha'(t) \cdot v = 0$$

for all t , so $\alpha(t) \cdot v$ is constant on I . We already know that $\alpha(0) \cdot v = 0$, hence $\alpha(t) \cdot v = 0$ for all t . $\square$

Problem. 1.2.5. Let $\alpha: I \rightarrow \mathbb{R}^3$ be a parametrized curve, with $\alpha'(t) \neq 0$ for all $t \in I$. Show that $|\alpha(t)|$ is a nonzero constant if and only if $\alpha(t)$ is orthogonal to $\alpha'(t)$ for all $t \in I$.

Solution. Of course, $|\alpha(t)|$ is a nonzero constant if and only if its square $|\alpha(t)|^2 = \alpha(t) \cdot \alpha(t)$ is as well. We have

$$\frac{d}{dt}|\alpha(t)|^2 = 2\alpha(t) \cdot \alpha'(t) = 0,$$

so $\alpha(t)$ and $\alpha'(t)$ are orthogonal if and only if $\alpha(t)$ is nonzero and $t \mapsto |\alpha(t)|^2$ is a constant function $I \rightarrow \mathbb{R}$, i.e. $|\alpha(t)|$ is a nonzero constant. $\square$

Problem. 1.3.5. Let $\alpha: (-1, +\infty) \rightarrow \mathbb{R}^2$ be given by

$$\alpha(t) = \left(\frac{3at}{1+t^3}, \frac{3at^2}{1+t^3} \right).$$

Prove that:

- (a) For $t = 0$, α is tangent to the x -axis.
- (b) As $t \rightarrow +\infty$, $\alpha(t) \rightarrow (0, 0)$ and $\alpha'(t) \rightarrow (0, 0)$.
- (c) Take the curve with the opposite orientation. Now, as $t \rightarrow -1$, the curve and its tangent approach the line $x + y + a = 0$.

Solution. (a) Of course, $\alpha(0) = (0, 0)$, so the curve intersects the x -axis at $t = 0$. To show that it's tangent to the x -axis, we can compute

$$\alpha'(t) = \left(\frac{3a(1+t^3) - 3at(3t^2)}{(1+t^3)^2}, \frac{6at(1+t^3) - 3at^2(3t^2)}{(1+t^3)^2} \right) = \left(\frac{3a - 6at^3}{(1+t^3)^2}, \frac{6at - 3at^4}{(1+t^3)^2} \right),$$

so that $\alpha'(0) = (3a, 0)$ is a vector along the x -axis.

(b) The coordinates of α and α' are all rational functions in t such that in each case the degree of the polynomial in the numerator is smaller than that in the denominator. This is enough to imply that each of the rational functions goes to zero as t grows arbitrarily large.

(c) As $t \rightarrow -1$, the curve gets closer to $x + y + a = 0$ in the sense that $x + y$ becomes arbitrarily close to $-a$. In particular, the sum of coordinates of $\alpha(t)$ is

$$\frac{3at(1+t)}{(1+t^3)} = \frac{3at}{1-t+t^2}$$

which as $t \rightarrow -1$ approaches $\frac{-3a}{3} = -a$. Moreover, the tangent approaches $x + y = -a$ in that the derivative becomes arbitrarily close to a scalar multiple of $(1, -1)$, so that the slope of the tangent line approaches that of $x + y = -a$. We can see this by noting that the sum of the coordinates of α' , which is the derivative of $\frac{3at}{1-t+t^2}$, is equal to

$$\frac{3a(1-t+t^2) - 3at(2t-1)}{(1-t+t^2)^2}$$

which, by setting $t = -1$, we can compute is equal to zero in the limit as $t \rightarrow -1$. $\square$

Problem. 1.3.6. Let $\alpha(t) = (ae^{bt} \cos t, ae^{bt} \sin t)$ for $t \in \mathbb{R}$ with a and b constants, $a > 0$, and $b < 0$ be a parametrized curve.

(a) Show that as $t \rightarrow +\infty$, $\alpha(t)$ approaches the origin 0, spiraling around it.

(b) Show that $\alpha'(t) \rightarrow (0, 0)$ as $t \rightarrow +\infty$ and that

$$\lim_{t \rightarrow \infty} \int_{t_0}^t |\alpha'(u)| \, du$$

is finite.

Solution. (a) Since $b < 0$,

$$\lim_{t \rightarrow \infty} e^{bt} = 0,$$

and $\cos t$ and $\sin t$ are bounded so $ae^{bt} \cos t$ and $ae^{bt} \sin t$ both approach zero as t goes to $+\infty$.

(b) The tangent vector is

$$\alpha'(t) = (abe^{bt} \cos t - ae^{bt} \sin t, abe^{bt} \sin t + ae^{bt} \cos t),$$

and we have already shown that each of the addends in each coordinate will approach zero as t grows large because the addends are all products of bounded expressions with e^{bt} ; hence $\alpha'(t)$ approaches $(0, 0)$ as $t \rightarrow +\infty$. Moreover, we can compute

$$\int_{t_0}^t |\alpha'(u)| \, du = \int_{t_0}^t \sqrt{a^2 e^{2bu} (\cos^2 u + \sin^2 u)} \, du = \int_{t_0}^t ae^{bu} \, du = \frac{a}{b} (e^{bt} - e^{bt_0}),$$

so, as $t \rightarrow +\infty$, the e^{bt} term disappears, leaving behind the finite quantity

$$\frac{-aeb^{t_0}}{b}$$

(note that this is positive because $b < 0$). $\square$

Problem. 1.3.10. Let $\alpha: I \rightarrow \mathbb{R}^3$ be a parametrized curve. Let $[a, b] \subseteq I$ and set $\alpha(a) = p$, $\alpha(b) = q$.

(a) Show that for any constant vector v , $|v| = 1$,

$$(q - p) \cdot v = \int_a^b \alpha'(t) \cdot v \, dt \leq \int_a^b |\alpha'(t)| \, dt.$$

(b) Set $v = \frac{q-p}{|q-p|}$ and show that

$$|\alpha(b) - \alpha(a)| \leq \int_a^b |\alpha'(t)| \, dt;$$

that is, the curve of shortest length from a to b is the straight line joining these points.

Solution. (a) The equality follows from the fundamental theorem of calculus, since, if $\alpha(t) = (x(t), y(t), z(t))$ and $v = (v_1, v_2, v_3)$, then

$$\begin{aligned} \int_a^b \alpha'(t) \cdot v \, dt &= \int_a^b x'(t)v_1 + y'(t)v_2 + z'(t)v_3 \, dt \\ &= (x(b)v_1 + y(b)v_2 + z(b)v_3) - (x(a)v_1 + y(a)v_2 + z(a)v_3) \\ &= p \cdot v - q \cdot v = (p - q) \cdot v. \end{aligned}$$

Then, the inequality follows from

$$\int_a^b \alpha'(t) \cdot v \, dt \leq \int_a^b |\alpha'(t) \cdot v| \, dt \leq \int_a^b |\alpha'(t)| \cdot |v| \, dt = \int_a^b |\alpha'(t)| \, dt,$$

where $|\alpha'(t) \cdot v| \leq |\alpha'(t)| \cdot |v|$ is an application of the Cauchy-Schwarz inequality and the last equality uses $|v| = 1$.

(b) Taking v to be the given expression shows that

$$(q - p) \cdot v = \frac{(q - p) \cdot (q - p)}{|q - p|} = \frac{|q - p|^2}{|q - p|} = |q - p| = |\alpha(b) - \alpha(a)|$$

can, as in the result of part (a), be bounded above by $\int_a^b |\alpha'(t)| \, dt$, which is what we wanted. $\square$

Problem. 1.4.5. Show that the equation of a plane passing through three noncollinear points p_1 , p_2 , and p_3 is given by

$$(p - p_1) \wedge (p - p_2) \cdot (p - p_3) = 0,$$

where p is an arbitrary point in the plane.

Solution. Recall that if three vectors a_1, a_2, a_3 are coplanar with the origin of $\mathbb{R}^3$, then $\det(a_1, a_2, a_3) = 0$. If p lies in the plane through p_1, p_2, p_3 , then $p_1 - p, p_2 - p$, and $p_3 - p$ are coplanar with $p - p$, since subtracting p is just a translation by the vector $-p$. We can also negate all the vectors without affecting coplanarity, since reflecting through the origin doesn't affect coplanarity, so actually $p - p_1, p - p_2$, and $p - p_3$ are coplanar with 0. To finish, we have

$$(p - p_1) \wedge (p - p_2) \cdot (p - p_3) = \det(p - p_1, p - p_2, p - p_3) = 0. \quad \square$$

Problem. 1.4.10. The natural orientation of $\mathbb{R}^2$ makes it possible to associate a sign to the area A of a parallelogram generated by two linearly independent vectors $u, v \in \mathbb{R}^2$. To do this, let $\{e_1, e_2\}$ be the natural ordered basis of $\mathbb{R}^2$, and write $u = u_1 e_1 + u_2 e_2$, $v = v_1 e_1 + v_2 e_2$. Observe the matrix relation

$$\begin{pmatrix} u \cdot u & u \cdot v \\ v \cdot u & v \cdot v \end{pmatrix} = \begin{pmatrix} u_1 & u_2 \\ v_1 & v_2 \end{pmatrix} \begin{pmatrix} u_1 & v_1 \\ u_2 & v_2 \end{pmatrix}$$

and conclude that

$$A^2 = \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix}^2.$$

Since the last determinant has the same sign as the basis $\{u, v\}$, we can say A is positive or negative according to whether the orientation of $\{u, v\}$ is positive or negative. This is called the **oriented area** in $\mathbb{R}^2$.

Solution. The square of the area is

$$\begin{vmatrix} u \cdot u & u \cdot v \\ v \cdot u & v \cdot v \end{vmatrix} = \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix} \begin{vmatrix} u_1 & v_1 \\ u_2 & v_2 \end{vmatrix} = \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix}^2,$$

noting that a matrix has the same determinant as its transpose. $\square$

Problem. 1.4.11(a). Show that the volume V of a parallelepiped generated by three linearly independent vectors $u, v, w \in \mathbb{R}^3$ is given by $V = |(u \wedge v) \cdot w|$, and introduce an **oriented volume** in $\mathbb{R}^3$.

Solution. The expression $|(u \wedge v) \cdot w|$ is equal to $|u \wedge v| \cdot |w| \cos \theta$, where θ is the angle formed by the vector w and the normal to the uv -plane (since $u \wedge v$ is normal to the uv -plane).

We know that the norm of $u \wedge v$ is the area of the parallelogram generated by u and v , and $|w| \cos \theta$ is the length of the projection of w on $u \wedge v$, i.e. the distance from w to the uv -plane, which is the height of the parallelepiped between the two faces that are parallel to the uv -plane. Therefore, the volume of the parallelepiped is the product of the base area and the height, i.e. $|(u \wedge v) \cdot w|$. $\square$